

Person Re-Identification

Topics in Computer Vision



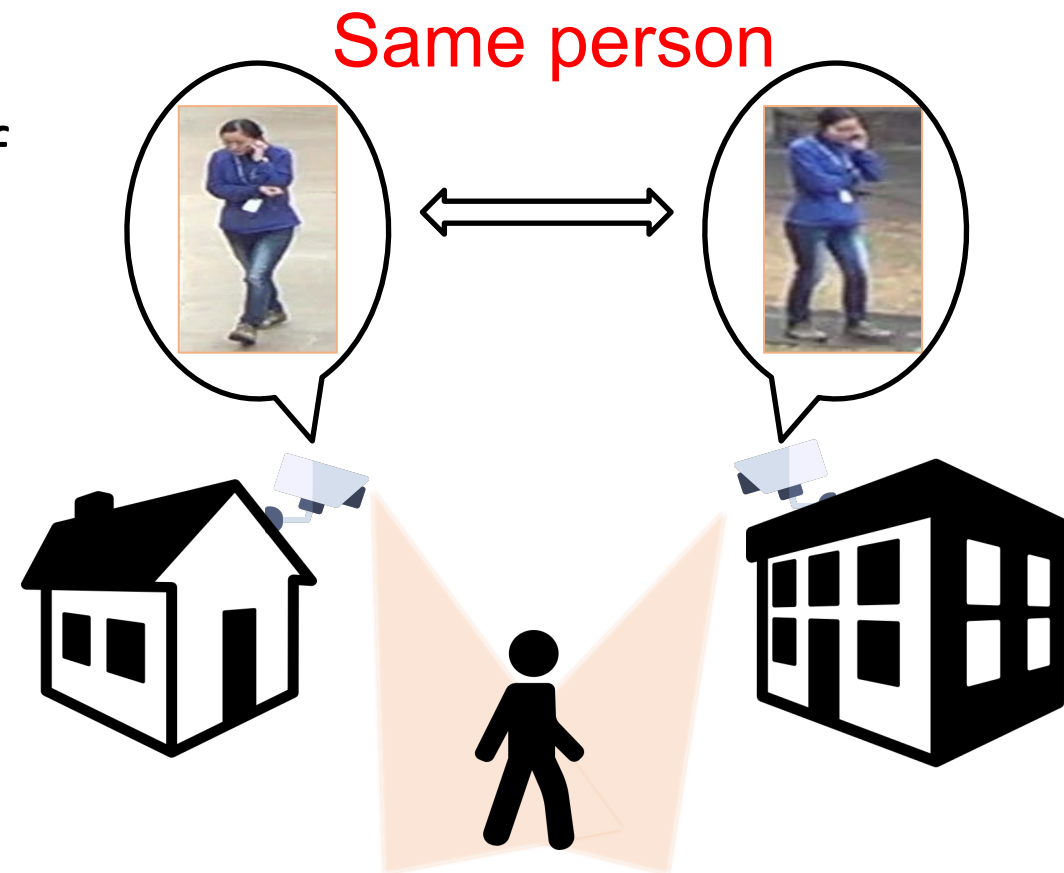
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What is Person Re-Identification

“Given an image/video of a person take from one camera, re-identification is the process of identifying the person from images/videos taken from a different camera with non-overlapping fields of views”

In other words, assigning a stable ID to a person in multi-camera setting



Challenges in Person Re-ID



Occlusion



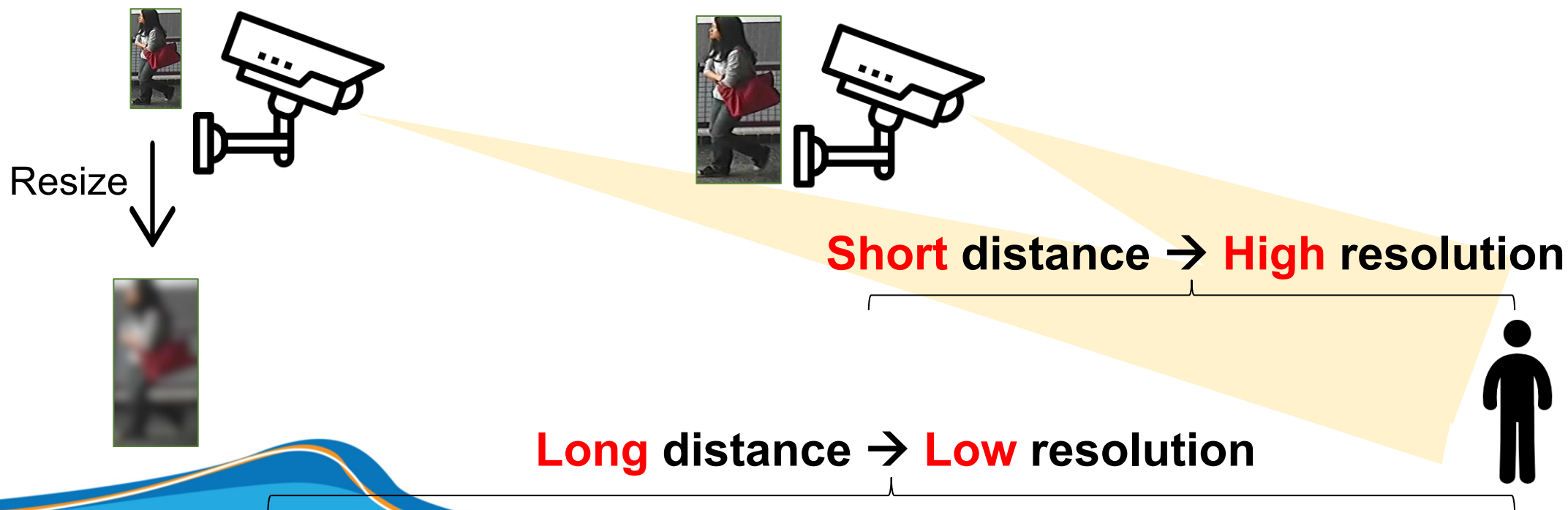
Background Clutters



Pose Variations

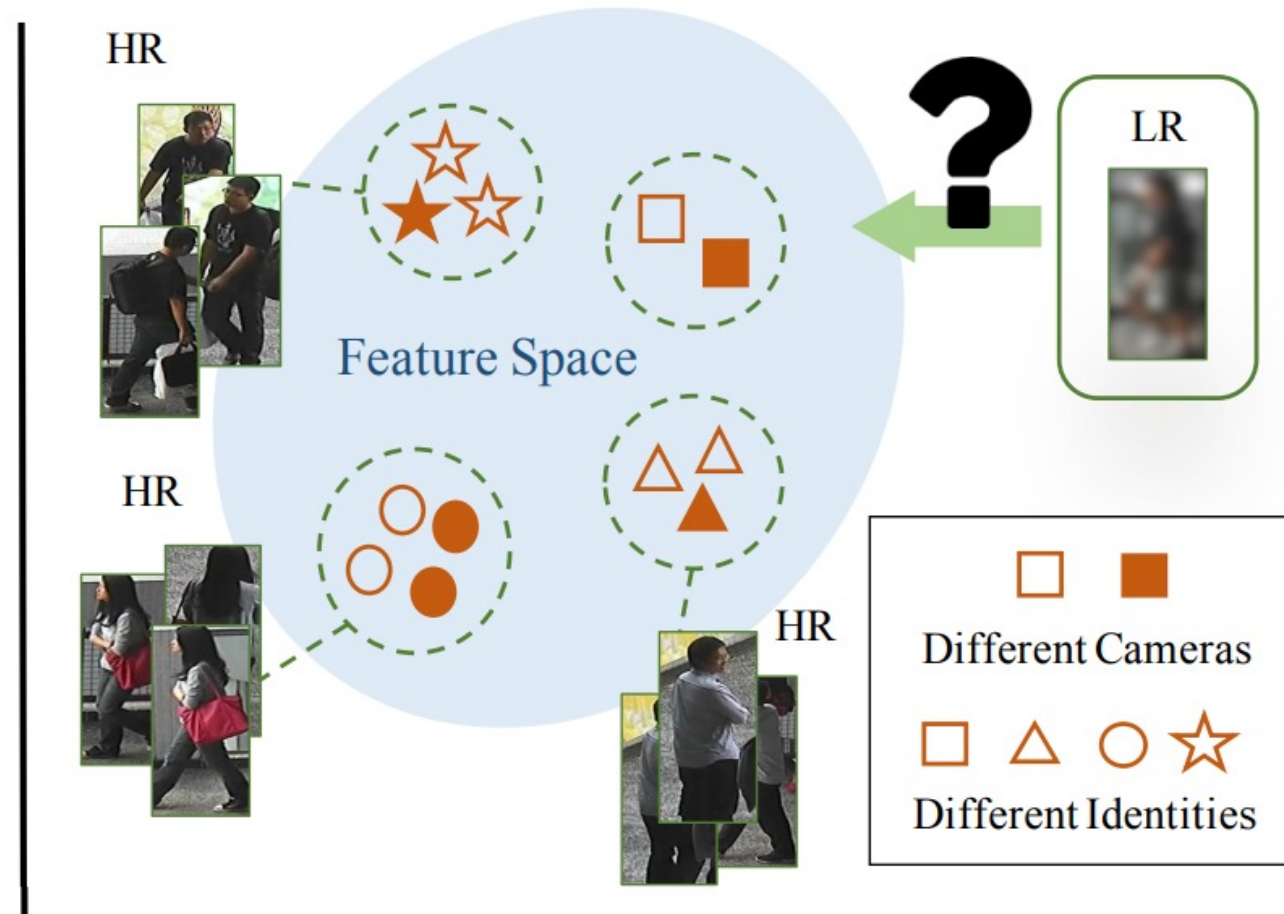
Resolution Variations

- Varying distances between the camera and the person of interest



Cross-Resolution Person Re-ID

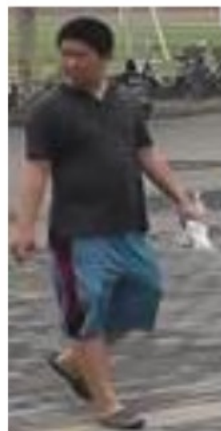
- Resolution mismatch issue due to cross-resolution images.



General Solutions

- Step 1: Feature Extraction
 - Extracting features for every person images
- Step 2: Feature Matching
 - Matching features to calculate the similarity score

General Solutions



Feature Extractor
(e.g. CNNs)

Image
Feature

Feature Extraction

Person
Image

Feature
Extractor

Feature

Person
Image

Feature
Extractor

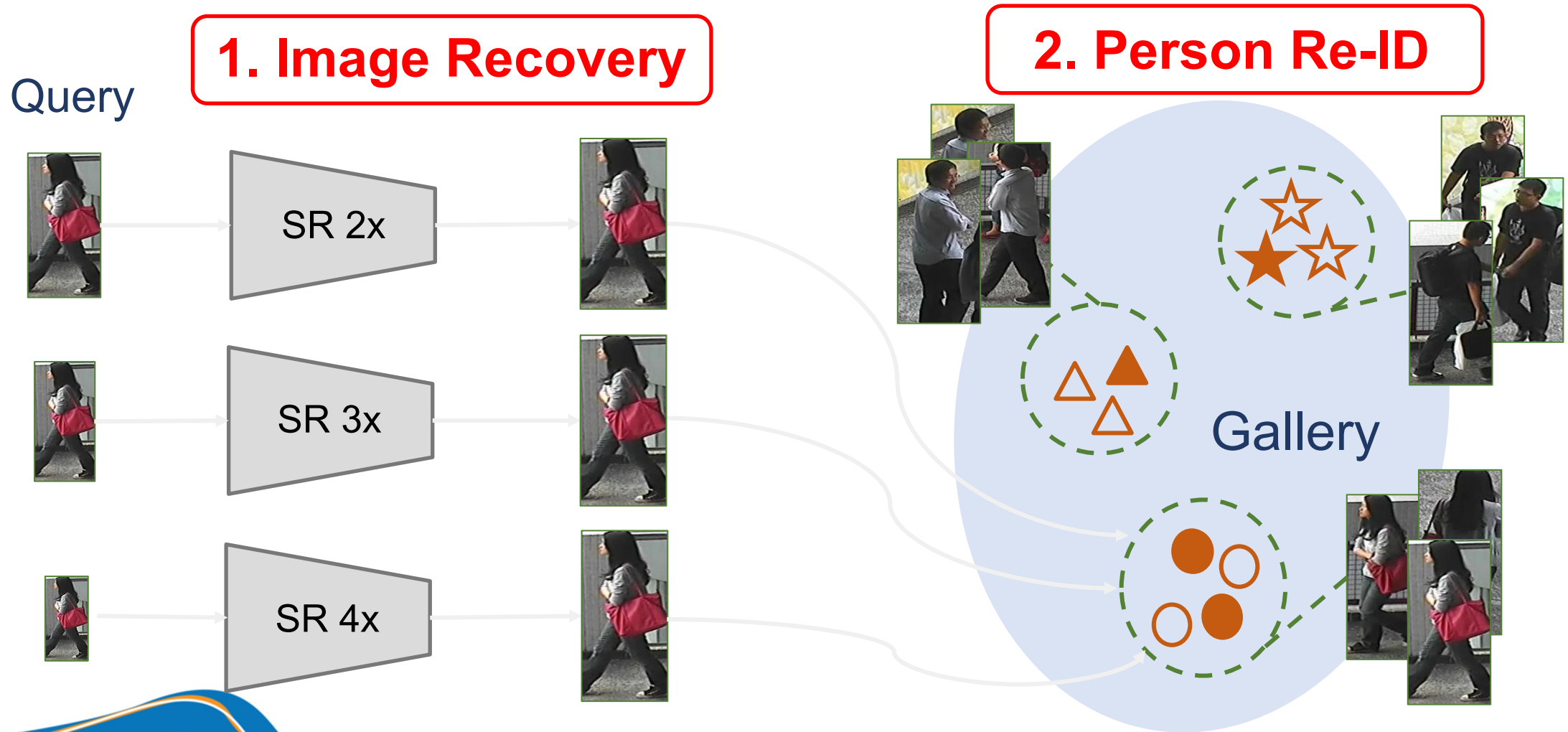
Feature

Feature
Matching

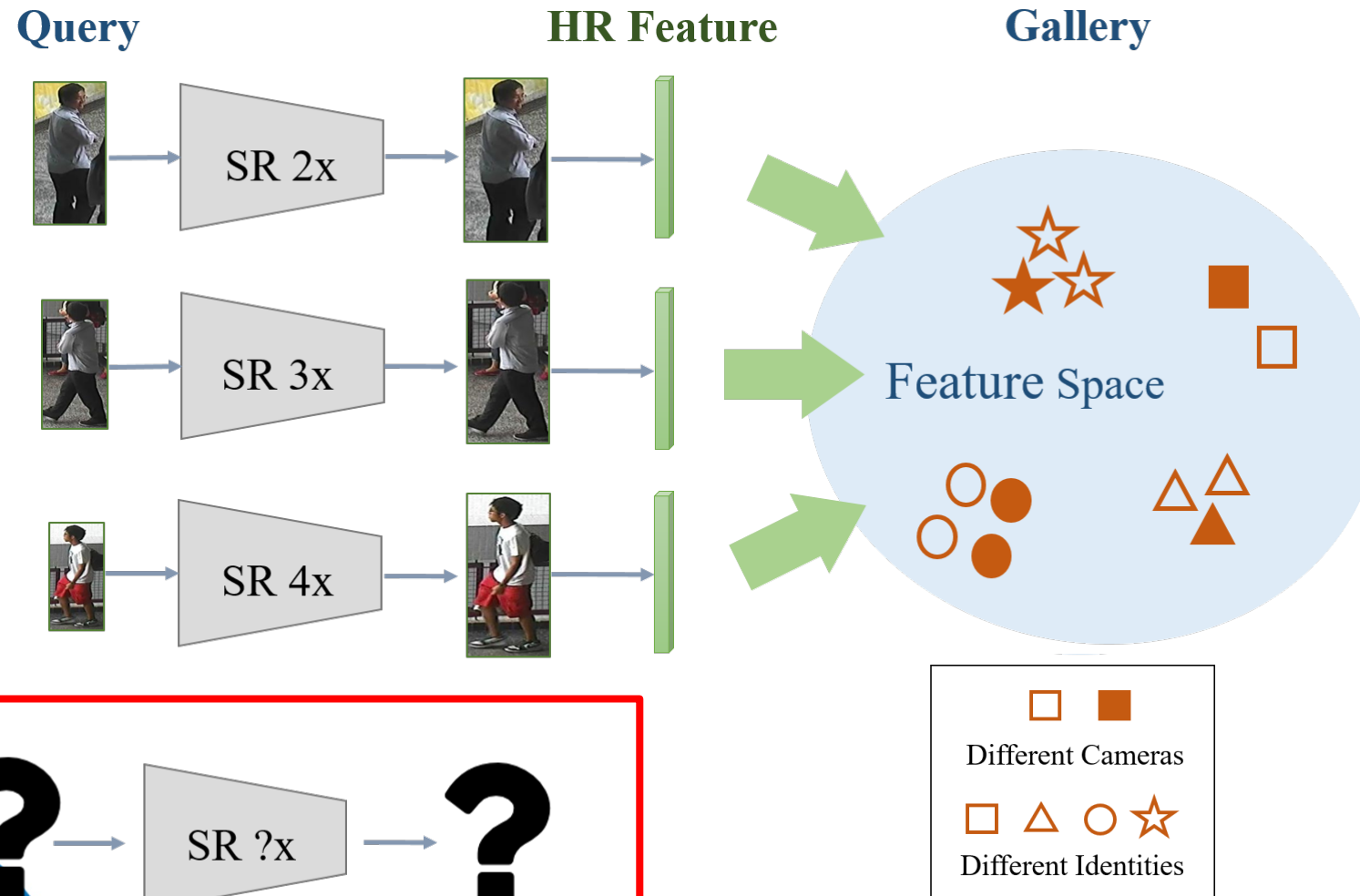
Feature Matching

Existing Strategies

Existing Strategies: SR for Cross Resolution Re-ID



Limitations of Existing Strategies



Unseen resolutions

Questions and Answers